

- (54) Which one of the following statements is incorrect? (a) If a chemical equation is written for a substance in its standard temperature and pressure conditions, then the substance is in its most stable state. (b) If a chemical equation is written for a substance in its standard temperature and pressure conditions, then the substance is in its least stable state. (c) The standard state of a dissolved substance is that in which its activity is unity. (d) The standard temperature of given substance is always 298,15 K.
- (55) Which of the following fulfils the conditions for balancing a thermochemical equation? (a) Balancing is done in such a way as only two moles of the substance are produced. (b) Balancing is done in such a way as only one mole of the substance is produced. (c) Balancing is done in such a way as no mole of the substance is produced. (d) Balancing is done in such a way as only one mole of the substance is not produced.
- (56) When water changes from ice to liquid water, which of the following enthalpies is involved? (a) Fusion (b) Fission (c) Vaporization (d) All of the above.
- (57) Calculate the pH of a 0.0020 M $\text{Ba}(\text{OH})_2$ solution. (a) 1.40 (b) 12.60 (c) 14.00 (d) 15.40
- (58) The pH of a solution increases from 1 to 2. The concentration of H^+ ions (a) decreases (b) increases (c) remains the same (d) becomes zero
- (59) The pH of a solution is 3, the hydrogen ion concentration in mole lit^{-1} is (a) 1×10^{-1} (b) 2×10^{-1} (c) 1×10^{-3} (d) 3×10^{-2}
- (60) The pOH of 0.1 M KOH is (a) 0.1 (b) 1.0 (c) -1.0 (d) 13
- (61) Which of these is NOT used in measuring pH? (a) pH meter (b) Litmus Paper (c) Acid - base Indicators (d) Starch Iodide Paper
- (62) The appropriate indicator for a weak acid - strong base titration is (a) Methyl red (b) Phenolphthalein (c) Methyl Orange (d) All of the above
- (63) Which of these is NOT a concept of acid and base (a) Arrhenius (b) Bronsted - Lowry (c) Le Chatelier's (d) Lewis
- (64) The appropriate indicator for a strong acid - strong base titration is (a) Methyl red (b) Phenolphthalein (c) Methyl Orange (d) All of the above
- (65) The amount of acid or base that may be added to the buffer solution before its pH changes appreciably is called (a) Acid - Base Strength (b) Acidic Strength (c) Buffer strength (d) Buffer capacity
- (66) Solutions that resist changes in pH on addition of little amount of acid or base are called (a) Indicator (b) Buffer (c) Amphiprotic (d) Amphoteric
- (67) All are methods of expressing rate of reaction EXCEPT: (a) initial rate (b) instantaneous rate (c) average rate (d) concentration rate.
- (68) Consider the reaction; $\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{I}^- \rightarrow 2\text{H}_2\text{O} + \text{I}_2$. If the average rate of formation of iodine in the above reaction is $10^{-6} \text{ mol/dm}^3/\text{sec}$ and it takes 16.67×10^{-2} minutes, find the change in concentration of the iodine. (a) 10^{-5} g/dm^3 (b) 10^{-5} mol/dm^3 (c) 10^{-5} mol/cm^3 (d) $10^{-5} \text{ mmol/cm}^3$
- (69) A catalyst speed up the rate of a chemical reaction by (a) lowering the activation energy (b) increasing the activation energy (c) maintaining a fixed activation energy (d) none of the above.
- (70) $[\text{A}]_0(2\text{K})^{-1}$ is the half - life of a: (a) first order kinetics (b) second order kinetics (c) zero order kinetics (d) None of the above.

$$\frac{\text{Rate}}{A^n}$$

$$\ln 2 = \frac{k}{A_0 - A} \times \frac{1}{2K}$$

$$K = \frac{\text{Rate}}{A^n}$$

$$10.002$$

$$2 \quad 2$$

$$\text{pOH} = 2.69$$

$$2.39$$

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Use the information below to answer questions 35, 36 and 37 respectively. From the rate of the reactions below:

$$0.001 = k(0.02)^y \times (0.02)^x \dots\dots\dots (i)$$

$$0.0005 = k(0.01)^y \times (0.02)^x \dots\dots\dots (ii)$$

$$0.002 = k(0.01)^y \times (0.04)^x \dots\dots\dots (iii)$$

(35) Calculate the order of the reaction with respect to y: (a) 2 (b) 3 (c) 1 (d) 4

(36) Find the value of x: (a) 2 (b) 3 (c) 4 (d) 1

(37) Determine the specific rate constant using equation (ii) above: (a) 125 mol²/sec (b) 125 mol/sec (c) 125 mol²dm⁶ (d) none of the above

(38) From graphical point of view, the rate of reaction is equal to the square of the reactants.

(39) A reaction in which three molecules take part in a chemical reaction is known as a tetramolecular reaction.

(40) A catalyst is a substance which speeds up the rate of a reaction but does not get affected by the process.

(41) In the standard notation for a voltaic cell, the double vertical line "||" represents: (a) a phase boundary (b) gas electrode (c) a wire (metal) connection (d) a salt bridge.

(42) Which of the following statements is FALSE? (a) Oxidation and reduction half-reactions occur at electrodes in electrochemical cells. (b) All electrochemical reactions involve the transfer of electrons. (c) Reduction occurs at the cathode. (d) All voltaic (galvanic) cells involve the use of electricity to initiate non spontaneous chemical reactions.

(43) How much work is required to move 2.36 mmol of electrons through a potential difference of 1.05 V. (F = 9.649 X 10⁴ C/mol). (a) 239.1 J (b) 293.1 J (c) 239.8 J (d) 249.4 J.

(44) What is the number of moles of silver formed when a current of 0.3 A is passed through a Silver nitrate for 30 min? (F = 96500) (a) 5.6 x 10⁻² mol (b) 5.3 x 10⁻³ mol (c) 5.6 x 10⁻³ mol (d) 6.5 x 10⁻³ mol.

(45) Find the mass of the product formed at the cathode when a dilute sulphuric acid solution is electrolyzed with a current of 0.6 A for 90 min. (F = 96500, H = 1.0). (a) 0.0336 g (b) 0.0168 g (c) 0.0618 g (d) 0.0363 g.

(46) A sample of gas occupies 12 L under a pressure of 1.2 atm. What would its volume be if the pressure were increased to 2.4 atm? (a) 6 L (b) 5 L (c) 0.167 L (d) 0.168 L.

(47) A sample of nitrogen occupies 117 cm³ at 100 °C. At what temperature in °C would it occupy 234 cm³ if the pressure did not change? (a) 473 °C (b) 746 °C (c) 0.00134 °C (d) 200 °C.

(48) A reaction vessel of 500 cm³ is filled with oxygen at 25 °C and the final pressure exerted by it is 101325 Nm⁻². How many moles of oxygen are there? (R = 8.314 JK⁻¹mol⁻¹).

(a) 0.02 mole (b) 20448.4 mole (c) 243745.5 mole (d) 0.20 mole. $PV = nRT$

(49) A small bubble rises from the bottom of a lake, where the temperature and pressure are 8 °C and 6.4 atm, to the water's surface, where the temperature is 25 °C and the pressure is 1.0 atm.

Calculate the final volume (in cm³) of the bubble if its initial volume was 2.1 cm³. (a) 14.2 cm³ (b) 0.0702 cm³ (c) 42 cm³ (d) 41.25 cm³.

(50) How many O₂ molecules will react with 25 CH₄ molecules according to the equation, CH₄ + 2O₂ → CO₂ + 2H₂O? (a) 5 (b) 25 (c) 30 (d) 50.

(51) Which of the following parameters directly affects the energy of given matter? (a) Temperature. (b) Free and directional properties. (c) Fixed and directional properties (c) None of the above.

(52) In which of the following states of matter are particles expected to have the highest energy? (a) Solid (b) Gas (c) Liquid (d) Magma or gel

(53) Which of the following forces hold particles together in a given matter? (a) Intermolecular. (b) Intramolecular. (c) Brownian. (d) Van der Waals.



- (16) A solution contains 126 g of HNO_3 per litre of solution. What is the normality of this solution to the nearest whole number? (a) 12 (b) 1 (c) 4 (d) 2
- (17) A 2.5 g/dm^3 solution of KOH contains how many mole per dm^3 ? ($K = 39.098$; $O = 15.999$; $H = 1.008$) (a) 4.60 mol (b) 0.446 mol (c) 0.0464 mol (d) 0.0446 mol
- (18) 15.6 - gram sample of C_6H_6 is mixed with excess HNO_3 . We isolate 18.0 grams of $C_6H_5NO_2$. What is the percent yield of $C_6H_5NO_2$ in this reaction? [$C_6H_6 = 78.1 \text{ g}$; $C_6H_5NO_2 = 123.1 \text{ g}$]. (a) 73.2 % (b) 0.73 % (c) 17.3 % (d) 1.72 %
- (19) According to the Kinetic Theory of matter, particles are said to be.... (a) Constant motion and random motion. (b) Free and directional motion (c) Fixed and directional motion (d) None of the above.
- (20) While atoms and molecules are used to describe the microscopic properties of matter, which of its properties do pressure and volume describe? (a) Nano-scale (b) Macroscopic (c) Mega-scale (d) None of the above
- (21) Stoichiometry of a chemical reaction refers to (a) the relative number of ions of substances involved in a chemical reaction. (b) the relative number of compounds of substances involved in a chemical reaction. (c) the relative number of electrons of substances involved in a chemical reaction. (d) the relative number of moles of substances involved in a chemical reaction.
- (22) What is the mass of 0.2 mole of calcium carbonate? (R.A.M.: $C = 12, O = 16, Ca = 40.1$). (a) 20.02 g/mol (b) 20.20 g (c) 20.02 g (d) 20.20 g/mol.
- (23) Calculate the number of gold atoms in a 20 g gold coin. ($Au = 197.0$, Avogadro's constant = 6.02×10^{23}). (a) 6.12×10^{22} (b) 6.10×10^{22} (c) 6.11×10^{22} (d) 6.01×10^{22} .
- (24) A magnesium chloride solution contains 10 g of magnesium chloride. Calculate the number of chloride ions in the solution. ($Mg = 24.3$, $Cl = 35.5$). (a) 1.263×10^{23} (b) 1.264×10^{23} (c) 1.260×10^{23} (d) 1.246×10^{23} .
- (25) How many moles of Na^+ ions are present in 20 cm^3 of a 0.45 M $NaCl$ solution? (a) $9 \times 10^{-2} \text{ mol}$ (b) $9 \times 10^{-3} \text{ mol}$ (c) $7 \times 10^{-3} \text{ mol}$ (d) $8 \times 10^{-3} \text{ mol}$.
- (26) What is the mass of a carbon dioxide molecule? ($C = 12$, $O = 16$). (a) $7.30 \times 10^{-23} \text{ g}$ (b) $7.31 \times 10^{-23} \text{ g}$ (c) $7.32 \times 10^{-23} \text{ g}$ (d) $7.13 \times 10^{-23} \text{ g}$.

Use the following information to answer questions 27 and 28: Given 6 moles of water ($H = 1$, $O = 16$, Avogadro's constant = 6.02×10^{23}):

- (27) How many molecules are there? (a) 3.612×10^{24} (b) 3.61×10^{24} (c) 3.621×10^{24} (d) 3.613×10^{24} .
- (28) How many atoms are there? (a) 1.084×10^{24} (b) 1.084×10^{25} (c) 1.083×10^{25} (d) 1.083×10^{24} .
- (29) The molar volume of nitrogen gas is $24 \text{ dm}^3 \text{ mol}^{-1}$ at RTP. Find the density of nitrogen gas ($N = 14$). (a) 1.617 g/dm^3 (b) 1.167 g/dm^3 (c) 0.167 g/dm^3 (d) 1.168 g/dm^3 .
- (30) In voltaic cells, the salt bridge (a) is not necessary for the cell to work. (b) acts as a mechanism to allow mechanical mixing of the solutions. (c) acts as a liquid wire allowing ions to flow and complete the circuit. (d) is tightly plugged with firm agar gel through which ions cannot pass.
- (31) The rate of a reaction is represented as $\text{Rate} = K[A][B]^3[C]^2[D]^4$ when the concentrations are in mmol/dm^3 and the time in seconds, find the overall order of the reaction. (a) 7 (b) 9 (c) 8 (d) 10
- (32) If the rate of a reaction is dependent on the concentration of the product, then the rate of the reaction will be: (a) negative (b) positive (c) neither positive nor negative (d) All of the above
- (33) If the rate of a reaction is plotted against $[A]^2$, then the slope of the graph is (a) gradient (b) 2.00 (c) specific rate constant (d) 3.00
- (34) $2H_2O_2 \rightarrow 2H_2O + O_2$. The most important factor that can affect the rate of the reaction is (a) concentration (b) catalyst (c) surface area (d) none of the above

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- (1) The strength of a base depends upon the (a) concentration of H^+ ions in aqueous solution (b) concentration of OH^- ions in aqueous solution (c) concentration of electron pair in aqueous solution (d) concentration of base.
- (2) The strength of an acid depends on the (a) number of H atoms present in its molecule (b) concentration of protons given by nit on ionization (c) concentration of water (d) concentration of acid
- (3) -If a solution has a pH of 5.50, calculate its $[OH^-]$ (a) 8.5 (b) 3×10^{-2} (c) 3.2×10^{-9} M (d) 3.2×10^{-9}
- (4) Consider the reaction, $2A + B \rightarrow A_2B$. Suppose a reversible reaction occurs by a one-step mechanism, the rate of the forward reaction is (a) $k_f[A]^2[B]$; (b) $k_f[A]_2[B]$; (c) $k_f[A_2B]$; (d) $k_f[A^2B]$
- (5) Consider, $W + X \rightleftharpoons Y + Z$. The rate of the forward reaction is proportional to the reactant concentrations raised to some (a) Power only (b) Power and coefficient (c) moles only (d) moles and molecules.
- (6) Consider the reaction, $A + B \rightleftharpoons C + D$, When we add more 'A' to an equilibrium mixture, the rate increases so that it no longer matches the rate of the ----- (a) Reverse reaction (b) Forward reaction (c) Forward and reverse reaction (d) None of the above
- (7) If a reactant or product is removed from a system at equilibrium, the reaction that produces that substance occurs more rapidly and to a greater extent than its ----- (a) Forward reaction (b) Reverse reaction (c) Forward and reverse reaction (d) All of the above
- (8) If $K > 1$, that means the reactant concentration is less than the product concentration. Hence there is ----- (a) More yield of the products (b) less yield of the products (c) more reactants present (d) less reactants present
- (9) Consider the reaction, $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$, $\Delta H = + 57.2$; which direction will the equilibrium shift if temperature is raised? (a) to the right (b) to the left (c) to the left and right (d) to the middle
- (10) Substances that undergo a definite colour change within a fairly narrow range of pH are called (a) Acid - base indicators (b) Acids (c) Amphiprotics (d) Bases
- (11) How many moles of water could be produced by the reaction of 35 mol of methane with excess oxygen in the following equation, $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$? (a) 0.7 mol (b) 7 mol (c) 17 mol (d) 70 mol.
- (12) What mass of CO_2 could be formed by the reaction of 16.0 g of CH_4 with 48.0 g of O_2 according to the equation, $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$? (a) 33 g (b) 44 g (c) 64 (d) 83 g.
- (13) Consider the reaction, $C_3H_{8(g)} + 5O_{2(g)} \rightarrow 3CO_{2(g)} + 4H_2O_{(g)}$
When 2.5 mol of O_2 are consumed in the reaction, what mol of CO_2 are produced?
(a) 1.5 mol (b) 2.5 mol (c) 5.0 mol (d) 6.0 mol
- (14) A 0.1014 g sample of purified glucose was burned in a C-H combustion train to produce 0.1486 g of CO_2 and 0.0609 g of H_2O . An elemental analysis showed that glucose contains only carbon, hydrogen, and oxygen. Determine the mass of oxygen in the sample. (C = 12.01, H = 1.008, O = 16). (a) 0.54 g (b) 0.054 g (c) 0.045 g (d) 0.053 g.
- (15) The equivalent weight of HNO_3 is.....(H=1.008 N=14.007 O=15.999). (a) 6.3012 g/equiv. (b) 6.30 g/equiv. (c) 63.0g/equiv. (d) 63.012g/equiv.

$$\begin{array}{ccccccc} 16 & 48 & - & x & & 0.1623 & \\ & 1 & & 2 & & 1 & 1 \end{array}$$

$$\frac{2.5}{160} = \frac{x}{132}$$